Rosen — §9.5: Equivalence Relations

Selected Exercises

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Exercise 9

Suppose that A is a nonempty set, and f is a function that has A as its domain. Let R be the relation on A consisting of all ordered pairs (x, y) such that $f(x) = f(y)$.

(a) Show that R is an equivalence relation on A.

(b) What are the equivalence classes of R?

Exercise 10

Suppose that A is a nonempty set and R is an equivalence relation on A. Show that there is a function f with A as its domain such that $(x, y) \in R$ if and only if $f(x) = f(y)$.

Exercise 15

Let R be the relation on the set of ordered pairs of positive integers such that $((a, b), (c, d)) \in R$ if and only if $a + d = b + c$. Show that R is an equivalence relation.

Exercise 16

Let R be the relation on the set of ordered pairs of positive integers such that $((a, b), (c, d)) \in R$ if and only if $ad = bc$. Show that R is an equivalence relation.

Exercise 25

Show that the relation R on the set of all bit strings such that $s\,R\,t$ if and only if s and t contain the same number of 1s is an equivalence relation.

Exercise 54

A partition P_1 is called a refinement *of the partition P_2 if every set in P_1 is a subset of one of the sets in P_2.*

Suppose that R_1 and R_2 are equivalence relations on a set A. Let P_1 and P_2 be the partitions that correspond to R_1 and R_2, respectively. Show that $R_1 \subseteq R_2$ if and only if P_1 is a refinement of P_2.

Exercise 56

Suppose that R_1 and R_2 are equivalence relations on the set S. Determine whether each of these combinations of R_1 and R_2 must be an equivalence relation.

(a) $R_1 \cup R_2$

(b) $R_1 \cap R_2$

(c) $R_1 \oplus R_2$

Exercise 58

Each bead on a bracelet with three beads is either red, white, or blue. Define the relation R between bracelets as: (B_1, B_2), where B_1 and B_2 are bracelets, belongs to R if and only if B_2 can be obtained from B_1 by rotating it or rotating it and then reflecting it.

(a) Show that R is an equivalence relation.

(b) What are the equivalence classes of R?

Exercise 60

(a) Let R be the relation on the set of functions from $\mathbb{Z}^+$ to $\mathbb{Z}^+$ such that (f, g) belongs to R if and only if f is $\Theta(g)$ (see Section 3.2). Show that R is an equivalence relation.

(b) Describe the equivalence class containing $f(n) = n^2$ for the equivalence relation of part (a).

Exercise 63

Do we necessarily get an equivalence relation when we form the transitive closure of the symmetric closure of the reflexive closure of a relation?

Exercise 68

Let $p(n)$ denote the number of different equivalence relations on a set with n elements (and, by Theorem 2, the number of partitions of a set with n elements). Show that $p(n)$ satisfies the recurrence relation

$$p(n) = \sum_{j=0}^{n-1} \binom{n-1}{j} p(n-j-1)$$

and the initial condition $p(0) = 1$. (*Note*: The numbers $p(n)$ are called *Bell numbers* after the American mathematician E. T. Bell.)